

AWS Glue

04-data-engineering/glue

Overview

Glue is AWS's serverless data integration service. It combines a data catalog, schema discovery, and ETL style jobs for moving and transforming data across analytical pipelines.

For ML, Glue is often the bridge between raw operational data and curated training or analytics datasets that downstream systems can trust.

Core Concepts

- The Glue Data Catalog stores table definitions and partitions so services like Athena and Spark jobs can agree on dataset structure.
- Crawlers inspect data sources and infer schemas, though mature teams often override or curate those schemas rather than trusting inferred output blindly.
- Glue jobs can run Spark based transformations at scale, with job bookmarks to track progress across repeated incremental runs.

How It Works

- Data lands in S3 or another source system, a crawler or manual schema registration defines metadata, and a Glue job transforms the data into a cleaned output location.
- The curated data is then queryable through Athena, consumable by SageMaker, or passed into downstream reporting and feature engineering systems.
- Schedules or event driven triggers let the pipeline refresh as new source data arrives.

AI and ML Relevance

- Glue helps standardize training data preparation, feature extraction jobs, and schema management for large datasets that exceed what ad hoc notebook code should be trusted to handle.
- It is especially useful when multiple consumers need the same curated datasets and you want transformation logic to live in a managed, repeatable pipeline.

Operational Notes

- Schema drift is common. If upstream producers change field names or types, downstream jobs can silently degrade unless validation is explicit.
- Partition data and use efficient file formats such as Parquet so jobs and queries do not waste time scanning unnecessary data.
- Glue is powerful, but not every transformation needs Spark. Use simpler tools for small, local, or latency sensitive tasks.

What To Remember

- Think of Glue as both metadata infrastructure and transformation infrastructure.
- The catalog is often just as important as the ETL job because discoverability and shared schema matter across teams.
- Reliable ML depends on reliable data preparation, and Glue is one way to formalize that layer.